5 Explanations based on Estimation Error

- Explanations based on estimation error argue that the **'normal' return was not correctly measured**, creating the **'appearance of an abnormal return'**.
- The two key ingredients of the asset pricing models are not observable:
  - **Betas** may be incorrectly measured and may be time-varying:
    - *Characteristics-scaled betas*: $\beta_{it} \sim Z_{it}$—Avramov and Chordia (2006).
    - *Stochastic betas*: $\beta_{it} \sim \beta_{it-1}, \nu_{it}$—Jostova and Philipov (2005).
  - **Risk premium** may be incorrectly measured and may be time-varying:
    - The key is the distinction between *realized* and *expected* risk premium.
  - Moreover, if betas are incorrectly measured, **beta's true correlation with the market return will be incorrectly measured**. An incorrect correlation leads to an incorrect estimate of the mispricing (alpha):

$$\begin{aligned} E[r_{it}] &= E[\beta_{it} R_{m,t}] + \alpha_i \\ &= E[\beta_{it}]\ E[R_{m,t}] + cov(\beta_{it}, R_{m,t}) \end{aligned} \tag{110}$$

5.1 Models for betas

A. Characteristics-scaled betas:

– Avramov and Chordia (2006) allow beta to vary with firm characteristics (size, BM, and past returns) as well as with macroeconomic variables:

$$r_{it} = \alpha_i + \underbrace{\beta_i(\mathbf{z}_{t-1}, \mathbf{z}_{it-1})'}_{conditioning} f_t + \epsilon_{it} \tag{111}$$

where $\mathbf{z}_t$ is a macro-level conditioning information
and $\mathbf{z}_{it}$ denotes a firm-level characteristic, e.g. size, book-to-market.

– AC find that with constant betas, none of the models captures the three anomalies.

– When beta is allowed to vary, size and value effects are often explained, but the momentum anomaly remains robust.

– Take-away: characteristics-scaled betas explain some but not all of the anomalies examined.

B. Stochastic betas:

- Jostova and Philipov (2005) propose a mean-reverting stochastic beta model.
- **Intuition**: betas respond to macroeconomic and company-specific news. The *news arrive randomly* implying a stochastic beta process.
- The SBETA model is:

$$\begin{aligned} r_{pt} &= \beta_{pt} r_{mt} + \sigma_p \varepsilon_{pt}, & \varepsilon_{pt} &\sim N(0,1) \quad (112) \\ \beta_{pt} &= \alpha_p + \delta_p(\beta_{p,t-1} - \alpha_p) + \sigma_{\beta_p}\nu_{pt}, & \nu_{pt} &\sim N(0,1) \\ (\alpha_p, \delta_p, \sigma^2_{\beta_p}, \sigma^2_p) &\sim p(\alpha_p, \delta_p, \sigma^2_{\beta_p}, \sigma^2_p) \end{aligned}$$

  where r_{pt} and r_{mt} are excess returns, β_{pt} is portfolio p's beta, α_p, δ_p, and $\sigma^2_{\beta p}$, are the unconditional mean, persistence, and conditional volatility of β_p, σ^2_p is idiosyncratic volatility, and $p(\alpha_p, \delta_p, \sigma^2_{\beta_p}, \sigma^2_p)$ is the joint distribution of the parameters. ε_{pt} and ν_{pt} are the stochastic components of returns and betas.

- The SBETA model is estimated using **Bayesian methods**, which offer a number of desirable features. For one, Bayesian prior-posterior relations reflect the *investors' learning process*. A Bayesian approach also incorporates the inherent *uncertainty about model parameters*. Finally, a *Markov Chain Monte Carlo algorithm* provides exact finite sample

inference, unlike MLE or GMM, which run into convergence problems and rely on asymptotic approximations.

- **In a simulation study**, the SBETA model generates significantly **more precise beta estimates** than GARCH betas, betas conditioned on aggregate or firm-level variables, and rolling-regression betas, *even when the true betas are generated based on these competing specifications*.
- **In out-of-sample tests**, the model **generates superior beta forecasts**, which significantly improves out-of-sample hedging effectiveness, reducing the volatility of hedging errors relative to OLS by up to 30%.
- **In asset-pricing tests**, stochastic betas **provide stronger support for the conditional CAPM and help resolve asset-pricing anomalies** such as the size, book-to-market, and idiosyncratic volatility effects in the cross-section of stock returns. The asset-pricing tests are based on individual stocks and demonstrate a *significant market risk premium* and *no non-risk related firm-specific returns*. Specifically, SBETA betas explain 23% of the cross-sectional variation in individual stock returns, while OLS, GARCH, and conditional betas explain between one and six percent.

– The fact that the SBETA estimates help resolve previously documented asset-pricing anomalies lends empirical support to the theoretical predictions of Gomes, Kogan, and Zhang (2003) that firm-level size and book-to-market are associated with expected returns because they are correlated with the true beta.

Fig 2. OLS and SBETA estimates vs generated true betas.

500 data samples of return and beta series of 120 observations each are simulated under different parameter settings. The figure illustrates the simulated and estimated beta series in a randomly selected sample. The thick line represents the simulated true beta series assuming $\sigma_\beta = 0.25, \sigma = 0.01$, and α and δ given on the top of each plot. The thin continuous line gives the estimated posterior mean of the SBETA β_t series using 2,000 draws of the Gibbs sampler. The dotted line illustrates OLS betas, $\widehat{\beta}_t$, estimated in a 60-month rolling-regression using the simulated return series and the market returns from periods $t - 29$ to $t + 30$.

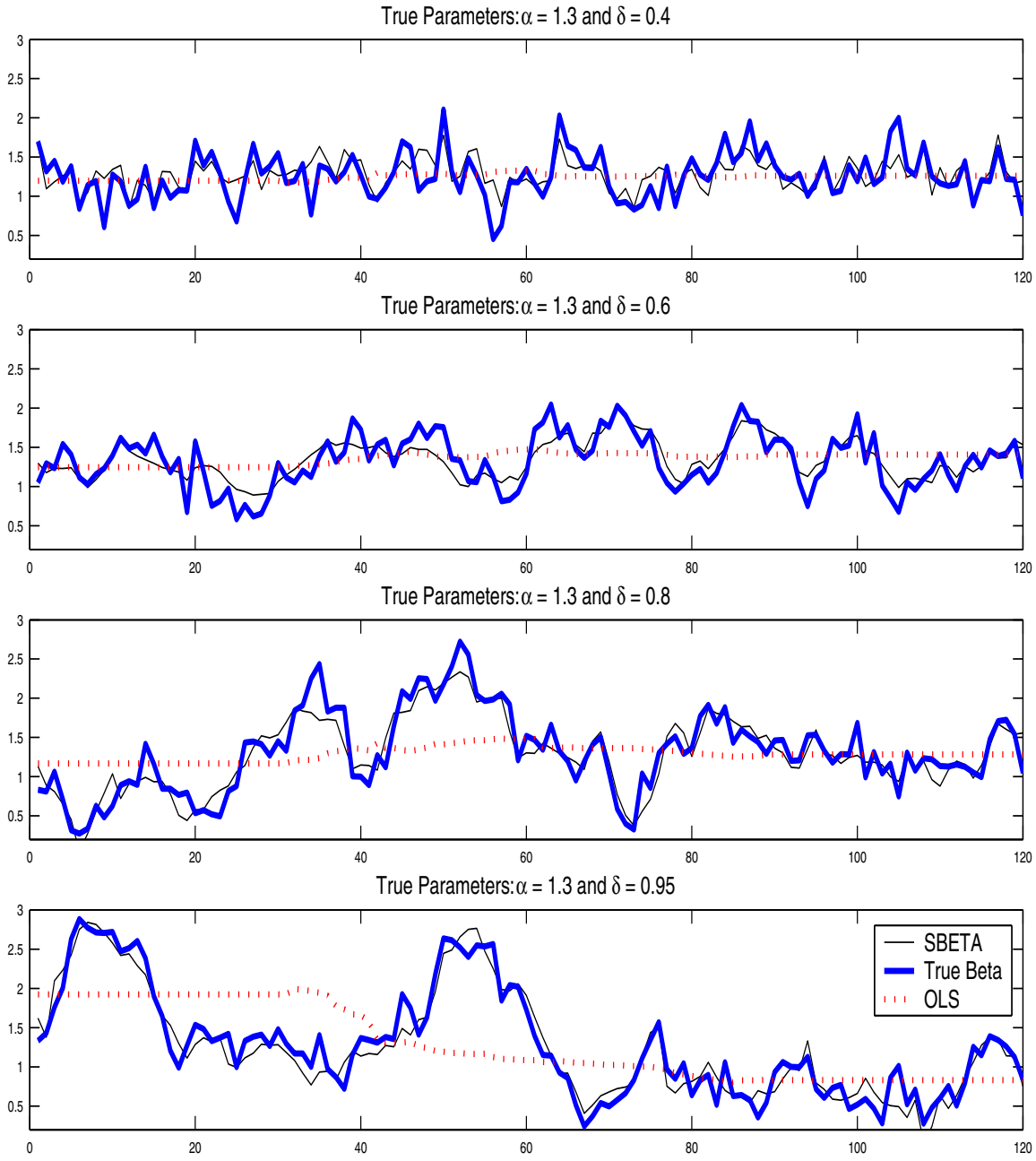
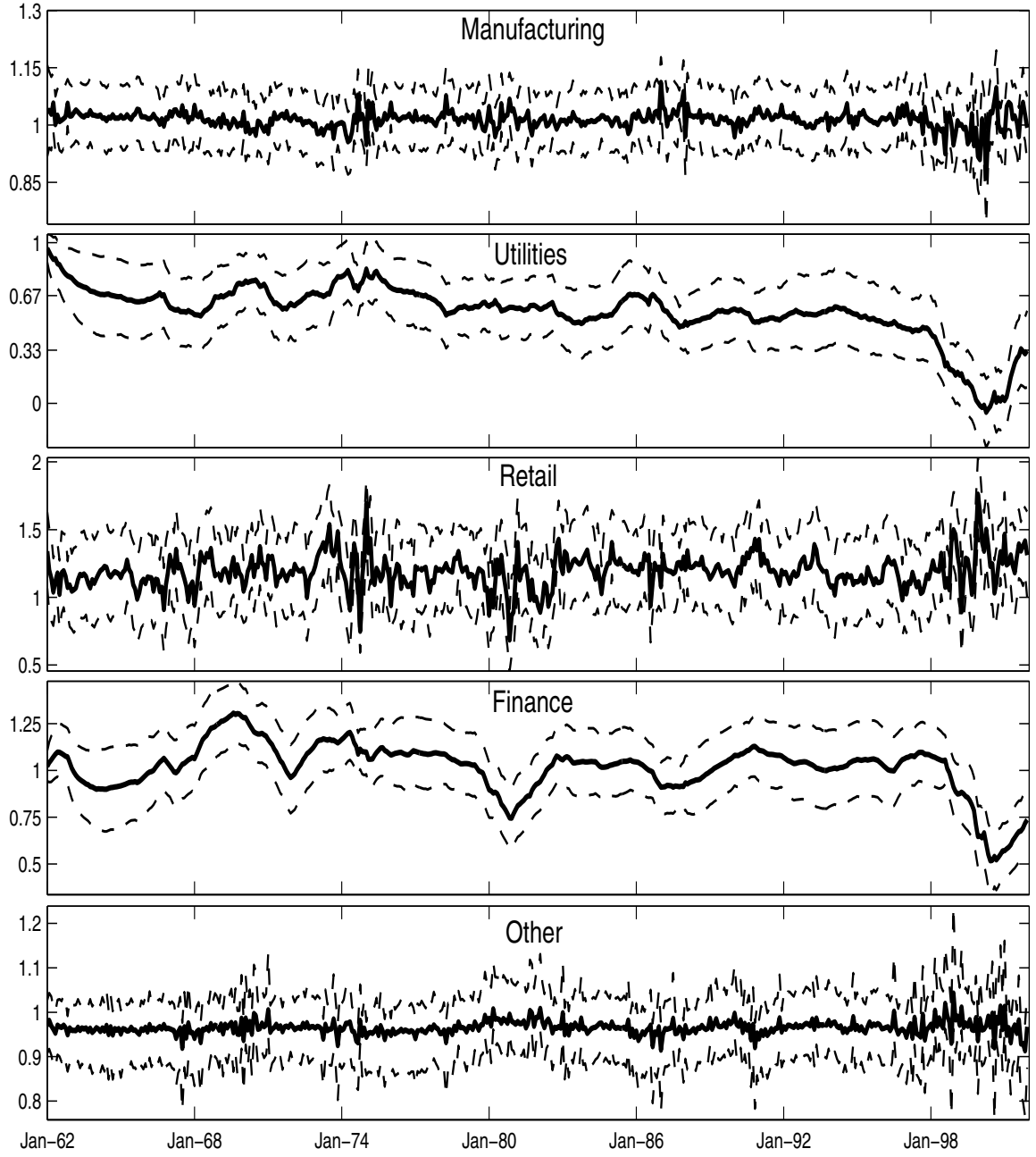


Fig 5. Estimated industry portfolio betas with 90% confidence bounds.

The SBETA model is fitted to industry portfolio returns over a 480-month period from January 1962 to December 2001. 45,000 draws are used to calculate the posterior means (after discarding the first 5,000 draws). The thick line in each plot illustrates the posterior mean of $\widehat{\beta}_t$, while the two thin dashed lines represent the 90% Bayesian posterior interval.



5.2 Models for risk premia

- **The Bloomberg model**: *analyst-based*
  - the **expected return for a stock** is based on the **dividend discount model**, which in turn *uses analysts' estimates* of betas, earnings per share forecasts, long term growth rate forecasts, and dividends.
  - the market expected return is the average of the stocks' expected returns.
- **The conditional market risk premium**: *instrument-based*
  - The risk premium is conditioned on economy-wide instruments. For example, Petkova and Zhang (2005) use the dividend yield (DIV), the default spread (DEF), the term spread (TERM), and the Treasury bill rate (TB).
  - The market risk premium is estimated in a time-series regression:

$$r_{mt+1} = \delta_0 + \delta_1\, DIV_t + \delta_2\, DEF_t + \delta_3\, TERM_t + \delta_4\, TB_t + e_{mt+1} \quad (113)$$

The conditional risk-premium is then:

$$\widehat{\gamma}_t = \widehat{\delta}_0 + \widehat{\delta}_1\, DIV_t + \widehat{\delta}_2\, DEF_t + \widehat{\delta}_3\, TERM_t + \widehat{\delta}_4\, TB_t \quad (114)$$

- NOTE: the Bloomberg model guarantees that the market risk premium is *positive*, while the conditional model does not.

5.3 Market Frictions

- **Market frictions** – focus primarily on why abnormal returns are not 'arbitraged away'. Do not necessarily explain how the abnormal return appeared in the first place.

5.4 Behavioral Biases

- **Behavioral biases** – explore biases in investor behavior that may give rise to deviations from 'rational' asset pricing.